# Sampling rate analysis:

## CG:5 Engine Specs:

impulse: 10 N·s
Burn time: 1.85 s
weight: 24.1 g
thrust max: 14.1 [N]

## 1:10 Nike smoke Rocket

Rocket weight = 68.04 g
= 0.06804 [kg]
Diameter = 41.66 mm

$m_{TOTAL} = 0.0241(kg) + 0.06804(kg)$

$m_{TOTAL} = 0.09214\ [kg]$

$F_g = m \cdot a = (0.09214\ kg)(9.81\ m/s^2)$

$F_g = 0.904\ [N]$

## FBD




Calculating $F_{T,avg}$:

$Impulse = 10\ N \cdot s = F_{T,AVG} \cdot t_{Burn} \rightarrow 10\ (N \cdot s) = F_{T,avg} \cdot 1.85\ (s)$

$\therefore F_{T,avg} = 5.4\ [N]$

Calculate $V_{max}$:

$V_{max} = V_0^{\,0} + a_{avg} \cdot t_{Burn}$

$V_{max} = 48.87\ [m/s^2] \cdot 1.85\ [s]$

$V_{max} = 90.41\ [m/s]$

$F_{Burn} = F_{T,avg} - F_g$

$= 5.4\ [N] - 0.904\ [N]$

$F_{Burn} = 4.496\ [N]$

$F = ma_{avg} \Rightarrow 4.496 = 0.092(a_{avg})$

$\therefore a_{avg,\,Burn} = 48.87\ m/s^2$

## Resolution calculations:

Sampling rate @ 60 Hz / 0.0167 s,
sampling rate @ 50 Hz / 0.02 s
Sampling rate @ 45 Hz / 0.02225,

## Δx between samples:

90.41 (0.0167) ≈ 1.5 m
90.41 (0.02) ≈ 1.81 m
90.41 (0.0222) ≈ 2 m

TAYLOR LANE

# Apogee calculations / time of operations

$$t_{total} = \underbrace{t_{pre\text{-}launch}}_{105} + \underbrace{t_{burn}}_{=1.85} + \underbrace{t_{coast\text{-}apogee}}_{=9.216\ (s)} + t_{descent}$$

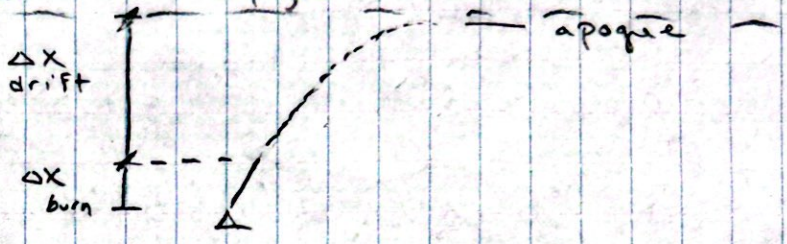
calculating $t_{coast\ apogee}$:

$$v_f = v_i + a \cdot t \rightarrow v_F = v_{max} + a \cdot t_{coast\text{-}apogee} \quad \left[ v_f = 0 \text{ at apogee},\ a = -9.81\ m/s^2 \right]$$

$$0 = 90.41 - 9.81\,(t_{coast\text{-}apogee}) \rightarrow t_{coast\text{-}apogee} = \boxed{9.216\ (s)}$$

calculating descent time:

$$\Delta x = v_i t + \frac{1}{2} a t^2$$




launch → burn

$$\Delta x_{burn} = \cancel{v_i^{0}} t + \frac{1}{2} a t^2$$

$$\Delta x_{burn} = \frac{1}{2}(48.87)(1.85^2) \rightarrow \Delta x_{burn} = 83.63\ [m]$$

end of burn to apogee

$$v_f^2 = v_i^2 + 2a\Delta x \quad (v_f = 0,\ v_i = 90.41,\ a = -9.81)$$

$$0 = (90.41)^2 + 2(-9.81)(\Delta x) \Rightarrow 0 = 90.41^2 - 19.62(\Delta x)$$

$$19.62(\Delta x) = 90.41^2 \rightarrow \Delta x_{drift} = 416.61\ m$$

$$\therefore \Delta x_{apogee} = 83.63\ [m] + 416.61\ [m] = 500.24\ m$$

assumes:
- rocket weight remains constant
- no air resistance
- no instability
- avg acceleration during Burn

factor in $F_{drag}$

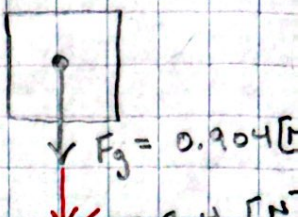
Factoring in drag Force post Engine Burn. Assuming

$\rho = 1.225\ kg/m^3$
$C_D = 0.5$
} initial analysis

$V_F = V_i + a(t_{coast\ apogee})$

$V_F$ → 0, $V_i$ = $V_{max}$

FBD:




→ Estimating drag Force:

$F = ma$ , $F_g = 0.904\ [N]$

$F_D = \frac{1}{2} \rho C_D A V^2$

$\rho = 1.225\ (kg/m^3)$ $A = \frac{1}{2}\pi D^2 = 0.0014\ [m^2]$

$C_D \approx 0.5$ , $F_D = \frac{1}{2}(1.225)(0.5)(0.0014)V^2$ $\Rightarrow$ $F_D = 0.0004174 V^2$

@ burn out, $V_{max} = 90.41\ (m/s)$

$\therefore F_{D,Burnout} = 3.4\ [N]$

$F = 0.904 + 3.4 = 4.304\ [N] = m \cdot a$

$\therefore a_{coast\ into\ Freefall} = 46.71\ (m/s^2)$ , $a_f = 9.81\ (m/s^2)$

drag varying linearly w/ velocity — [avg acceleration during coast: $\frac{-46.71 + -9.81}{2} = -28.26\ (m/s^2)$

apogee calculations: $\Delta x_{burn} = 83.63\ [m]$ — launch to End-of Burn

$V_F^2 = V_i^2 + 2a(\Delta x)$ ($V_F$ → 0)

$0 = (90.41^2) + 2(-28.26)(\Delta x)$ — End-of burn to apogee

$(56.52)(\Delta x) = 8173$

$\Delta x = 144.6\ [m]$ $\therefore$ apogee = 144.6 + 83.63 m

apogee = 228 [m]

$t_{total} = t_{pre\ launch} + t_{Burn} + t_{coast} + t_{descent}$

$t_{pre\ launch} = 10s.$ , $t_{Burn} = 1.853$ , $t_{coast} = 3.25$ , $t_{descent} = 54.8$

---

$\Delta x = \frac{v_i + v_f}{2} t$ ($v_f \to 0$)

$\Delta x = 228m - 83.63m = 144.6\ [m]$

$144.6\ (m) = \frac{90.41}{2} \cdot [t_{coast}] \longrightarrow t_{coast} = 3.2$ seconds

---

$t_{descent}$ calculations:

Parachute specs: $D_{para} = 0.381\ [m]$ , $C_D = 0.75$

$A_{para} = \pi r^2 = \pi \left(\frac{0.381}{2}\right)^2 \Rightarrow A_{para} = 0.114\ [m^2]$

@ terminal Velocity: $F_D = F_g$

$\frac{1}{2} \rho C_D A v^2 = mg \longrightarrow v = \sqrt{\frac{2mg}{\rho C_D A}}$

$v = \sqrt{\frac{2(0.09214)(9.81)}{(1.225)(0.75)(0.114)}} \longrightarrow v = 4.16\ m/s$ (speed of descent)

$v = \frac{x}{t} \longrightarrow 4.16 = \frac{228}{t_{descent}}$

$\therefore t_{descent} = 54.8\ [s] \longrightarrow t_{total} = 69.85$ second

$t_{total} \approx 70$ seconds , 60 seconds for recovery